

LITTLE STREAM DETECTOR

LSD 3.0

Technical Documentation

Capabilities, operation and result interpretation

Property	Value
Release	3.0
Edition	User-focused final edition with native VVC support
Platform	Windows PowerShell 5.1 + .NET Framework
External multimedia tools	None required
Primary use	Native stream inspection and A/B comparison
Revision	25 September 2026

Purpose of this edition This revision documents native H.266/VVC two-pass analysis, VVC frame-level SliceQPY, I/P/B reporting, and pixel-format derivation from SPS chroma format and bit depth.

1. What LSD 3.0 Does

Little Stream Detector 3.0 is a native Windows media inspection application for examining video files without launching FFmpeg, FFprobe, MediaInfo or another external multimedia executable. It reads supported containers, identifies tracks, analyzes supported elementary streams and presents human-readable and structured results.

Typical uses Inspect a single file, compare an encode with a reference, verify stream structure after remuxing, examine bitrate behavior, compare frame-level quantizer distributions, and validate native VVC picture structure.

Core capabilities

- Open a video from the file picker or by drag and drop.
- Analyze container, track, codec, timing, bitrate, frame-level information and pixel format where supported.
- Keep the latest analysis as A Current and optionally compare it with an immutable B Reference.
- Display independent Summary A and Summary B reports, Streams data, JSON and a processing Log.
- Plot A/B bitrate profiles and show interval values in a mouse-hover tooltip.
- Display QP or codec-equivalent quantizer statistics in percentages or raw frame counts.
- Report unavailable information as N/A rather than inventing or estimating a value.
- Allow a running operation to be cancelled.

Supported families

Area	Supported families
Containers	Matroska, WebM, AVI 1.0, OpenDML AVI 2.0, MP4, M4V, MOV
Video	H.264/AVC, H.265/HEVC, H.266/VVC, AV1, VP8, VP9, MPEG-4 Part 2/XviD
Audio	AAC, MP3, PCM, AC-3, E-AC-3
Outputs	Summary A/B, Streams, JSON, Log, bitrate profile, quantizer distribution

2. Operating the Application

Basic workflow

1. Choose Open video or drop a supported media file into the drop area.
2. Wait for analysis to complete. The current result becomes A Current.
3. Review Summary A, Streams, JSON and Log as needed.
4. Inspect the bitrate profile and quantizer panel.
5. To compare files, retain a result as B Reference and analyze the other file as A Current.
6. Use Remove Ref B when the reference is no longer needed.
7. Use Copy to copy the active report.

Main controls

Control	Purpose
Open video	Selects a file and starts analysis.
Drop video here	Accepts a file by drag and drop.
Cancel	Stops an analysis that is currently running.
Copy	Copies the currently active report.
Set Ref B / Remove Ref B	Stores or clears the comparison reference.

A/B comparison model

- A Current is the most recently analyzed result.
- B Reference is a stored comparison result that remains unchanged while another file is analyzed.
- The interface supports one current result and one reference result at a time.
- Blue represents A Current. Gold represents B Reference.

Comparison tip Use percentages when files have different frame counts or durations. Use raw counts when investigating the exact number of frames at a quantizer value.

3. Understanding the Reports

View	What it provides
Summary A	Readable report for the current file.
Summary B	Readable report for the stored reference.
Streams	Structured information about native tracks.
JSON	Serializable metadata and stream information.
Log	Processing lifecycle, validation notes and first-failure messages.

Information commonly shown

- File name, modification time, size and duration.
- Detected container and native document type.
- Number and type of native tracks.
- Video dimensions, codec family, pixel format and codec-specific configuration.
- Audio format, rates, channels and bitrate where available.
- Canonical analysis status and routing information.
- Frame types, keyframe information and codec-dependent quantizer statistics where analysis is complete.

Pixel format and bit depth

For native VVC streams, LSD derives pixel format from SPS chroma_format_idc and SPS bit depth. The result is displayed in common planar notation such as yuv420p, yuv420p10le, yuv422p10le or yuv444p12le. Monochrome streams use gray or a depth-qualified gray format.

SPS chroma format	Displayed family
0	gray / grayNle
1	yuv420p / yuv420pNle
2	yuv422p / yuv422pNle
3	yuv444p / yuv444pNle

Color metadata remains separate Pixel format does not imply color primaries, transfer characteristics, matrix coefficients or full/limited range. LSD reports those fields only when reliably signaled.

4. Bitrate Profile

The bitrate profile plots the amount of video data over presentation-time intervals. A Current is blue and B Reference is gold. This makes local peaks, quiet sections and differences between two files visible over the full duration.

Reading the graph

- Tall bars indicate intervals containing more encoded data.
- Low bars indicate intervals containing less encoded data.
- Peak bucket reports the largest interval value observed for the active file.
- The horizontal position corresponds to playback time.
- Hovering the graph displays the selected time interval and available A/B bitrate values.

What the graph can reveal

- Bitrate spikes around complex scenes or scene changes.
- Different bitrate allocation between an encode and its reference.
- Unexpectedly flat, sparse or irregular sections.
- Broad similarity after a lossless remux, allowing for documented timing or container differences.

Do not judge quality from bitrate alone A higher instantaneous or average bitrate does not automatically mean higher visual quality. Codec family, encoder settings, source complexity and quantizer behavior must be considered together.

5. QP and Quantizer Distribution

The quantizer panel summarizes the codec-specific quantizer metric for analyzed frames. AVC, HEVC and VVC use frame-level SliceQPY when complete native parsing succeeds. Other codecs use the corresponding quantizer index.

Codec family	Histogram value
H.264 / AVC	Frame-level SliceQPY
H.265 / HEVC	Frame-level SliceQPY
H.266 / VVC	Frame-level SliceQPY
AV1	Base Q Index
VP8	Base Q Index
VP9	Base Q Index
MPEG-4 Part 2	VOP quantizer

Statistics above the distribution

Statistic	Meaning
Frames	Number of frames contributing to quantizer analysis.
Avg	Arithmetic mean of analyzed values.
Median	Middle quantizer value after sorting.
Mode	Most frequently occurring quantizer value.
Range	Lowest and highest recorded values.

Percentage and count modes

Click the fixed QP distribution header to switch A/B columns between percentage mode (A % and B %) and count mode (A pcs and B pcs). The colored bars remain a relative visual representation and A/B columns stay aligned.

6. Interpreting Quantizer Results

Quantizer distributions are most useful as patterns, not as isolated numbers. A shift toward lower or higher values can be significant, but the meaning depends on codec family, encoder behavior, content and other settings.

Practical interpretation

- A narrow distribution indicates many frames were encoded within a relatively small quantizer range.
- A broad distribution indicates stronger variation in quantization across the file.
- A different mode shows that the most common frame quantizer changed.
- Different percentages with similar totals indicate a genuine change in distribution shape.
- Similar percentages with different raw counts can occur when files have different analyzed frame totals.
- For an unchanged elementary stream after a lossless remux, frame counts and quantizer histograms should normally remain equivalent.

Important comparison rule A and B distributions are directly comparable only when both files use the same quantizer metric. Different codec families are not a one-to-one quality scale.

What QP does not prove by itself

- QP alone does not measure visible quality.
- QP alone does not establish encoder efficiency.
- The same numeric value can have different practical meaning across codecs or encoding structures.
- LSD does not decode pixels and does not calculate PSNR, SSIM or VMAF.

Relative Quantization Profile

When valid quantizer data are available for both A Current and B Reference, clicking the upper chart switches between the bitrate timeline and a relative quantization profile. The native quantizer panel on the right remains unchanged.

Horizontal	Percentile of analyzed frame values, from 0 to 100%.
Vertical	Position of the value within the native quantizer scale of that codec, normalized to 0 to 100%.

Compatible metrics. The relative view can place widely separated distributions in one overview without replacing the exact native values or requiring the full native range to be inspected by scrolling.

Different metrics. Metrics such as AV1 Base Q Index and VVC SliceQP cannot be compared on one absolute QP axis. The relative view provides an approximate common context for distribution position and shape.

Interpretation rule. A crossing of the two curves means that both distributions reach the same relative position in their own codec-local scales at that percentile. It does not mean equal quantization step, equal compression efficiency, or equal visual quality. LSD does not decode pixels and does not calculate PSNR, SSIM or VMAF.

7. Codec and Audio Analysis

Video information

Family	Native analysis
H.264 / AVC	SPS/PPS/VUI, I/P/B, pixel format and SliceQPY
H.265 / HEVC	VPS/SPS/PPS, I/P/B, pixel format and SliceQPY
H.266 / VVC	vvvC, SPS/PPS, two-pass NAL and embedded-PH traversal, I/P/B, pixel format and SliceQPY
AV1	OBV traversal, frame state, geometry, pixel format and Base Q Index
VP8	Frame tag, KEY/INTER and Base Q Index
VP9	Superframes, frame state, references and Base Q Index
MPEG-4 Part 2 / XviD	VOL/VOP, I/P/B/S, GMC and VOP quantizer

Native VVC analysis

- Pass 1 indexes bounded MP4 samples and length-prefixed VVC NAL units.
- vvvC SPS/PPS context is validated before picture-level output is accepted.
- Pass 2 resolves one picture event per sample and traverses embedded Picture Headers and Slice Header prefixes.
- I/P/B counts and the SliceQPY histogram are published only when all VCL pictures are parsed and zero QP headers are rejected.
- Pixel format is derived from SpsChromaFormat and SpsBitDepth after successful VVC analysis.

Audio information

Family	Validation and reporting
AAC	Configuration, core/output rate, channels, SBR, PS, payload and bitrate
MP3	Version/layer, bitrate, sample rate, padding and frame size
AC-3 / E-AC-3	Syncword, bitstream ID, rate, frame size, blocks and channels
PCM	Block alignment, sample format, duration and bitrate

8. Support Matrix and Limitations

Video	Matroska/WebM	MP4/MOV	fMP4	AVI
H.264 / AVC	Yes	Yes	Yes*	Yes, Annex B
H.265 / HEVC	Yes	Yes	Yes*	No
H.266 / VVC	Yes**	Yes	Yes*	No
AV1	Yes	Yes	Yes*	No
VP8	Yes	No	No	No
VP9	Yes	Yes, vp09	Yes, vp09*	No
MPEG-4 Part 2	Yes	Yes	Yes*	Yes

* Fragmented MP4 reuses the existing codec analyzer when the sample entry and payload format are supported.

** VVC requires a supported length-prefixed NAL representation and valid codec-private context.

Known limitations

- Encrypted CENC samples and auxiliary encryption structures are not supported.
- Standalone media segments without an initialization moov box are not supported.
- Files still being written and live network streams are not supported.
- VP8 is supported only in Matroska/WebM.
- VP9 in AVI is not supported.
- AVI H.264/AVC support is intended for Annex B streams.
- AVI files containing AAC or E-AC-3 audio are not currently supported.
- Container-only color metadata is not used as a fallback.
- VVC subpicture, encrypted, unsupported partition, selected weighted-prediction and uncommon extension paths may be rejected rather than guessed.
- LSD does not decode picture pixels or calculate objective quality scores.

9. Validation, Troubleshooting and Good Practice

When an analysis is incomplete

8. Open Log and identify the first validation or parser failure.
9. Confirm that the file is complete and readable from local storage.
10. Check whether the container/codecs combination is supported.
11. Compare analyzed frame count with expected duration and frame rate.
12. Treat disabled or N/A histograms as unavailable rather than zero-valued results.
13. For VVC, check structural validation, PH prefixes, SH RPL prefixes, and SliceQPY parsed/rejected counts.

Useful expectations for a lossless remux

Check	Expected relationship
Video/audio sample count	Equal, or differences explained by container edits.
Payload bytes	Equivalent elementary-stream accounting.
Keyframe and I/P/B counts	Equal for unchanged elementary streams.
SliceQPY histogram	Equal for unchanged AVC/HEVC/VVC elementary streams.
Pixel format and bit depth	Equal when codec SPS information is unchanged.
Timestamps	Equivalent after documented edit-list effects.
Rejected samples/headers	Zero for a validated asset.

Best practice Keep source and comparison files clearly identified. Compare like with like, start with the same codec metric, and evaluate bitrate, frame counts, pixel format and quantizer distribution together.

Appendix A. Concise Technical Reference

LSD uses PowerShell for orchestration and the WinForms interface, with embedded C# for native parsing and analysis. Supported containers are converted into a shared, bounded sample model before codec and audio analysis. Standard MP4 and fragmented MP4 use separate indexing routes but feed the same reporting, bitrate and A/B comparison layers.

Processing flow

Pipeline File selection -> container detection -> sample indexing -> canonical media model -> codec/audio analysis -> validation -> reports and A/B graphs

Defensive behavior

- Sample ranges must remain inside the source file.
- Unavailable information is reported as N/A.
- Quantizer histograms remain unavailable when parser completeness checks fail.
- VVC I/P/B, SliceQPY and derived pixel format are published only from validated SPS/PPS and complete picture traversal.
- Fragmented MP4 support is limited to standard unencrypted files with initialization metadata and supported media fragments.

Terminology

Term	Definition
A Current	Most recently analyzed result.
B Reference	Stored immutable comparison result.
Canonical sample	Container-neutral bounded sample with timing and key state.
QP / quantizer distribution	Frequency distribution of the codec-specific quantizer metric.
Pixel format	Planar sample representation derived from chroma subsampling and bit depth, for example yuv420p10le.
VVC two-pass analysis	NAL indexing followed by context-aware picture and Slice Header traversal.
pcs mode	Raw per-quantizer frame counts for A and B.
fMP4	Fragmented MP4 whose media samples are described by movie fragments.